

A city is a mathematical object

Question: How much information forces one arrangement? Children progress from a complete list to a uniqueness proof to a lower bound on clues. These are questions about permutations, Latin squares, and determining sets.

Preparation and prerequisites

Use snap cubes: six per youngest child, eighteen per middle child, forty for the upper child: **112 cubes**, plus a few for the launch. Prebuild towers if time is short. Numeral slips can record a second solution after the physical attempt. Print three K-1 packets, three middle packets, one upper, one optional extra, and this guide. Build on the tabletop or quick scrap-paper grids with one-inch cells. The smaller printed grids record solutions; they are not sized to hold every brand of cube.

K-1: compare heights/count to 3; no independent reading; recognize different orders. Middle: count to 3 and follow row/column rules; adult may read; distinguish an example from a complete list. Upper: count to 4; track constraints, cases, and uniqueness; no formal algebra. Extra: understand a 4-by-4 Latin square, coordinate labels, and a pigeonhole argument.

Launch

Let towers be handled first. Show 2,1,3 from the left. Ask which towers a tiny person sees; touch each visible tower. Walk to the other end. Only after this introduce the separate city rule: every row and column uses each height once. A clue counts visible towers, not the first tower's height or the number of cubes.

A shared hour with seven children and two adults

0-10: explore materials freely. 10-15: brief launch and rules. 15-35: main investigation. 35-40: movement/reset; preserve work. 40-55: continue or choose an optional proof. 55-60: share and tidy. This flexible default is our proposal, not a source prescription.

One adult stays with K,K,1. The organizer alternates middle and upper, leaving a concrete next attempt and returning for an explanation. In games keep two opposing sides and rotate a checker. Proof continuations need adult attention; pages are not a completion quota.

K-1 stop: Build all six skylines and describe their views. **Optional proof:** organize by first tower to show completeness; explain the impossible one-at-each-end view.

Middle stop: Build both top-clue cities and select one with the second clue. **Optional proof:** certify the list, then swap unseen rows/columns to rule out every one-clue puzzle.

Upper stop: Explain the complete five-clue case table. **Optional proof:** check all five deletion certificates and prove the three-clue alternative forces the same city. The computer's minimum claim is separate.

Complete lists and minimum clues

K-1: six skylines, not six unrelated puzzles

The complete list (left count, right count) is: 123 (3,1), 132 (2,2), 213 (2,1), 231 (2,2), 312 (1,2), 321 (1,3). Fix the first tower: each of three choices leaves two orders. This proves completeness without factorial notation. One visible tower from both ends is impossible with distinct heights: the tallest would have to occupy both ends. Both 132 and 231 show two from each end.

The two 2-by-2 cities are 12/21 and 21/12. The top-row left clue 2 chooses the former. If row/column rules are too much, keep investigating rows; a complete skyline list is already substantial mathematics.

Middle: exactly two cities, then exactly one

Top clue 3 forces first column 1,2,3. Top row is either 123 or 132. Column exclusions force the respective completions:

1	2	3
2	3	1
3	1	2

1	3	2
2	1	3
3	2	1

The right clue 2 on row 2 keeps the first. Deleting that clue leaves both. Deleting only the top clue permits 123/231/312 and 312/231/123 (swap rows 1 and 3).

A genuine minimum proof: No single visibility clue can determine an entire 3-by-3 Latin square. If it sees a row, swap the two other rows; if it sees a column, swap the two other columns. The clue stays true and the square changes. Two clues suffice in our example, so the minimum over all uniquely solvable 3-by-3 visibility puzzles is two.

Hints and evidence

K: fix the first tower and exchange the other two. Middle: point down the first column, then try both top rows. Ask what rules out another answer. Accept a complete physical case split; long written prose is unnecessary. If children only solve the city, ask for one clue deletion and an alternative, saving the minimum proof for conversation.

A human certificate for five clues

Upper: the unique solution

Both the opening and the five-clue puzzle have rows **1234/2341/3412/4123**. Here is a small exhaustive argument for five clues. Bottom row starts with 4 and has right visibility 2, so 3 must be last. Only 4123 and 4213 remain. Keep both: the bottom clues alone do not rule either out; row 3 and row 2 do that next.

Row 3 has two views of 2. Its six possible rows are **1423, 2143, 2413, 3142, 3241, 3412**. Cross out candidates conflicting with the bottom row. The surviving pairs and possible row 2s are:

Bottom row	Row 3	Row 2 with right view 2
4123	3241	none
4123	3412	2341
4213	3142	none

Check the final column by listing row permutations avoiding the two entries already in each column, then testing the right view. The top row is forced by missing column heights. This proves uniqueness without trusting a solver.

Deleting clues: witnesses

Deleting L3, L4, R2, R3, R4 gives **2, 2, 6, 12, 20** solutions respectively. L3 means the left clue on row 3. One alternative per deletion proves necessity:

Delete	Alternative city, top row first
L3	1234 / 3412 / 2341 / 4123
L4	1234 / 4123 / 3412 / 2341
R2	1324 / 2431 / 3142 / 4213
R3	1324 / 3142 / 2431 / 4213
R4	1234 / 2143 / 3412 / 4321

On optional upper page 3, check each certificate is Latin, keeps all four retained clues, and differs from the target. Restoring the clue before the next deletion keeps the five arguments separate. The five-clue set is **minimal**, but it is **not minimum**: left clues 4,3,2 on rows 1,2,3 uniquely determine this same target. First row is 1234. The rows with left view 3 are 1243,1324,1342,2134,2314,2341; all but 2341 repeat a column entry. With rows 1 and 2 fixed, the possible row 3s are 3412 and 4123; only 3412 has left view 2. The last row is forced.

The script exhaustively verifies that three is the minimum *for this target city among subsets of its 16 perimeter clues*. That computation is not a proof that every 4-by-4 city needs three. For children, the concrete three-clue alternative already disproves the claim that minimal means minimum.

Trades, bounds, and research

Extra: a lower bound with an escape move

Each bold block has form ab/ba . Exchanging a, b inside it preserves the multiset in each affected row and column. With at most three entry clues, one of four disjoint blocks has no clue. Switch it: a distinct Latin square keeps every given entry. Thus at least four *entry* clues are necessary for this square.

Hitting these four blocks is not sufficient. Clues at $(1,1), (1,3), (3,1), (3,3)$ leave rows 2 and 4 unclued; swapping those rows keeps all four clues. Any uniquely determining set must meet *every* trade. Conversely, the changed cells in any alternative completion form a trade; hitting every such trade rules out alternatives.

Keep clue types separate: Previous pages use outside visibility clues; the extra uses fixed entries. Bounds do not transfer automatically.

Where the mathematics goes

Undergraduate combinatorics studies permutations, Latin squares, constraint satisfaction, and hitting sets. Joy Morris's *Combinatorics*, chapter on Latin squares develops Latin-square structure; our visibility questions add a different constraint system.

Hatami and Qian, *Teaching dimension, VC dimension and critical sets for Latin squares* (2016), §1 and Theorem 1.2, study sets of *entries* determining a square and prove a quadratic lower bound for sufficiently large orders. Our trade argument is a small instance of the question, not their asymptotic theorem. The paper states a sharper conjecture; no current open-status claim is needed here.

Teaching sources and returning children

JRMF *Skyscrapers Teaching Guide*, PDF pp. 1–4: physical viewpoints, city rules, and reasoning; pp. 13–14: blank boards. Our clue instances and investigations extend that format. *Math Circle by the Bay*, printed pp. ix–x: manipulatives, extra challenges, and themes at different depths. Lowell Handout 2.1–2.2 seating constraints are related prior material, not evidence children have solved these cities.

Use F05-K-v2/M-v2/U-v2/X-v2. Prepared, not recorded as taught. Save trade classification, Latin transversals, and larger skyline recurrences for later years. Rebuild with `build.sh`; `plans/verify-week-05.py` checks the finite claims independently of the explanations.